

Benchmark #6

3 MeV H on Fe

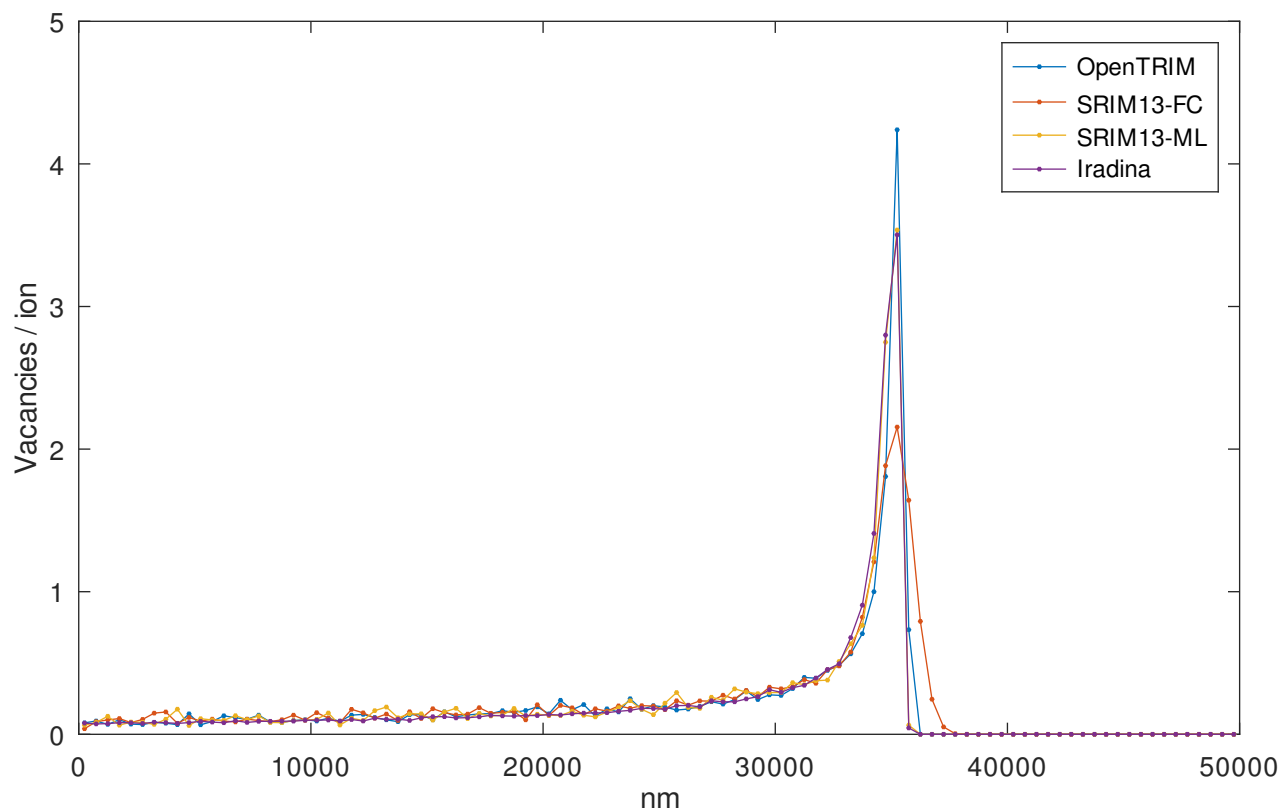
Ion energy E0 = 3e+06 eV

Target depth = 50000 nm

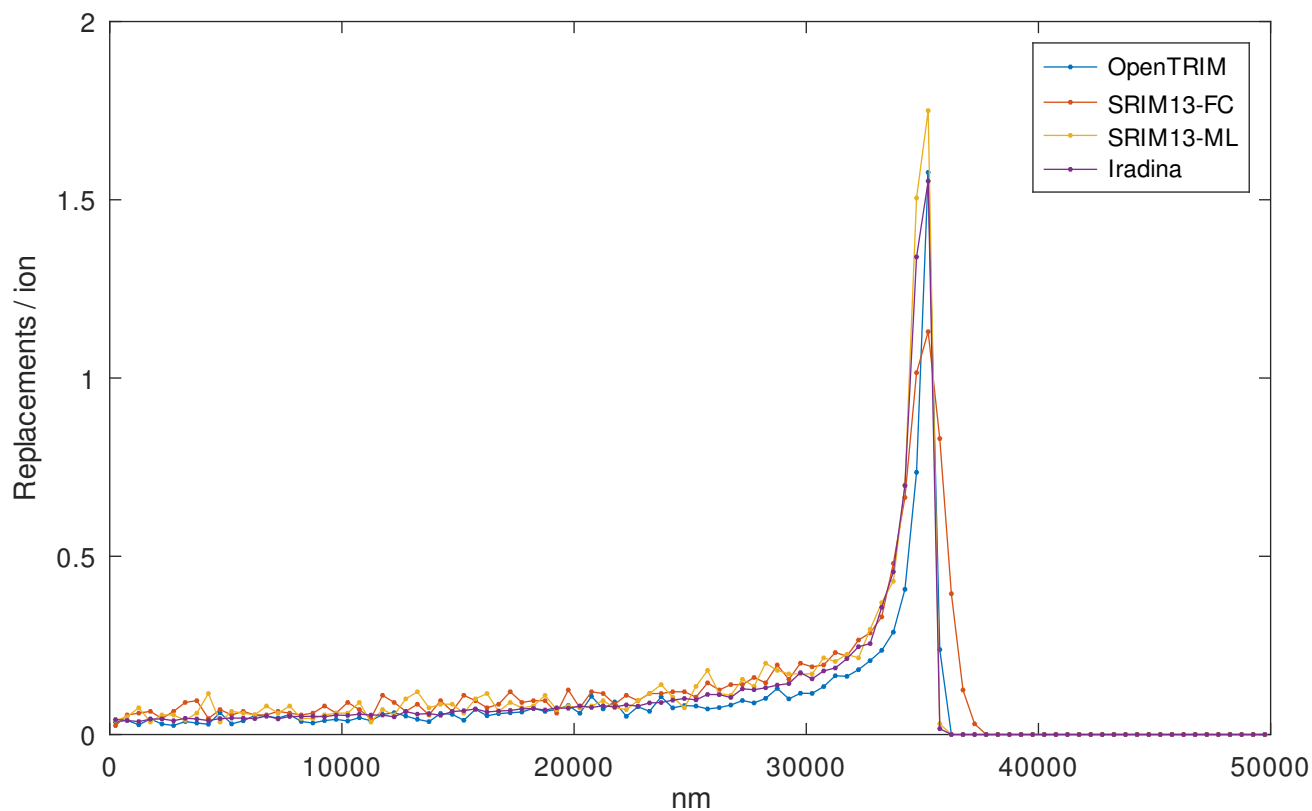
Summary Table

Quantity	OpenTRIM	SRIM13-FC	SRIM13-ML	Iradina
V(Fe)	20	20.9	20	19.5
R(tot)	8.08	11.9	11.4	9.95
I(H)	1	1	1	1
$\Delta E_e(H)/E_0$	0.999	0.999	0.999	0
$\Delta E_e(r)/E_0$	7.48e-05	7.53e-05	7.14e-05	0
$\Delta E_e(tot)/E_0$	0.999	0.999	0.999	0.999
$\Delta E_{ph}(H)/E_0$	0.000364	0.000316	0.000486	0
$\Delta E_{ph}(r)/E_0$	0.00054	0.000848	0.000994	0
$\Delta E_{ph}(tot)/E_0$	0.000904	0.00116	0.00148	0.000976
$1 - (\Delta E_e + \Delta E_{ph})/E_0$	0	-0.000302	-0.000382	-4.77e-07

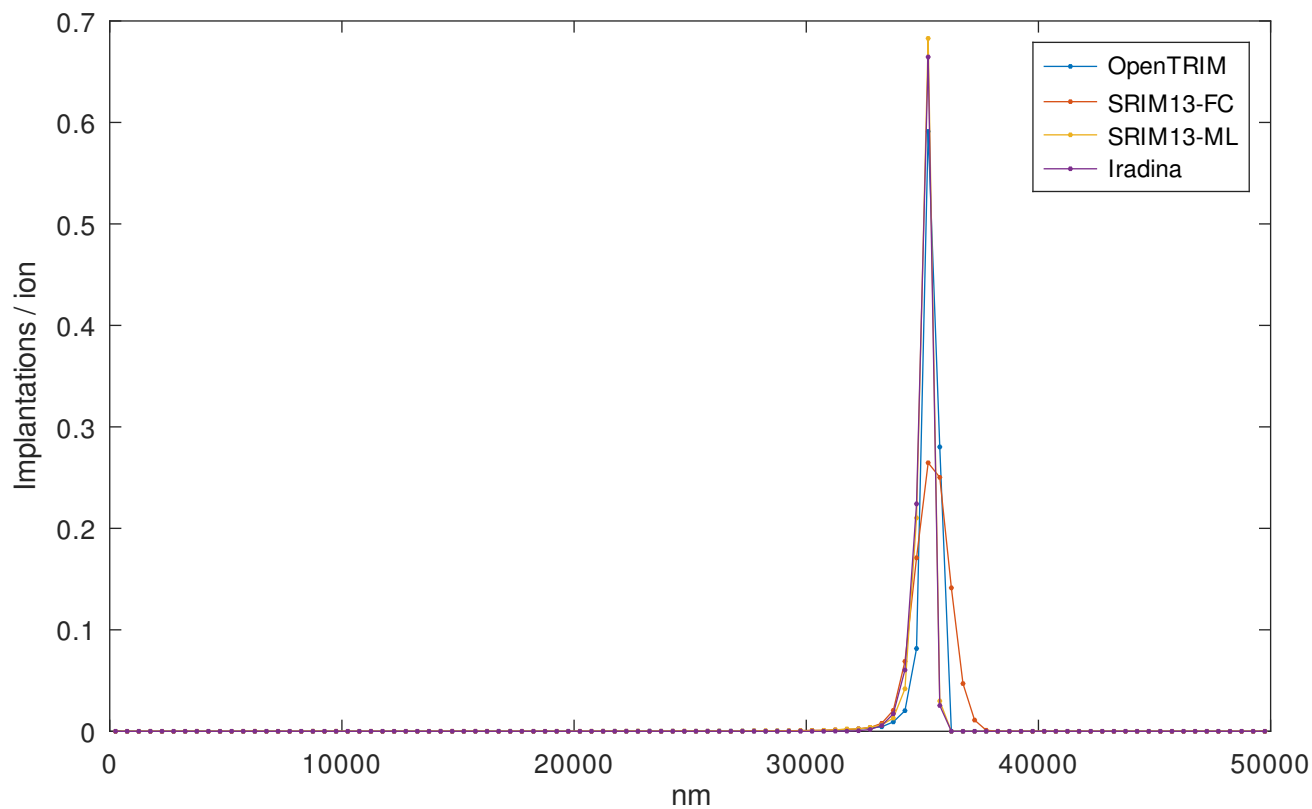
Vacancies of Fe in Fe



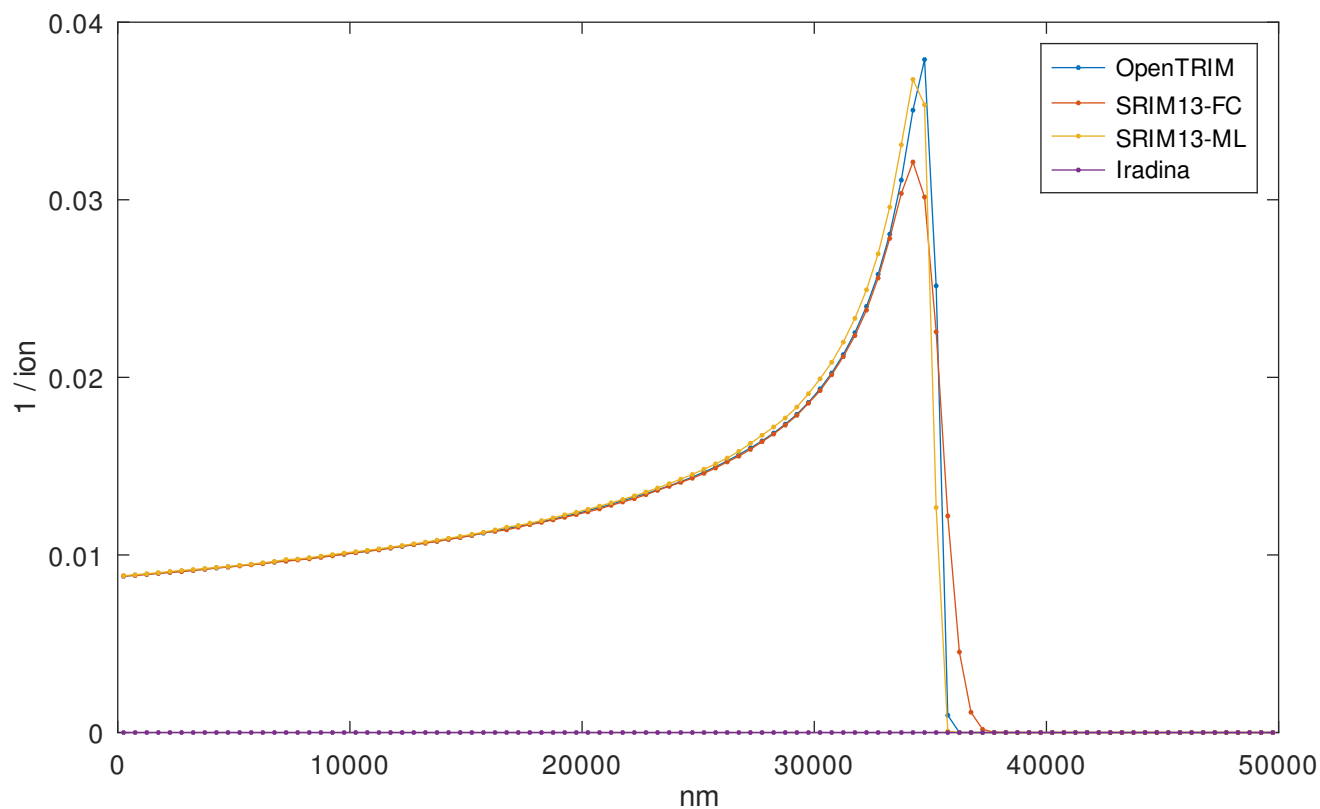
Replacements



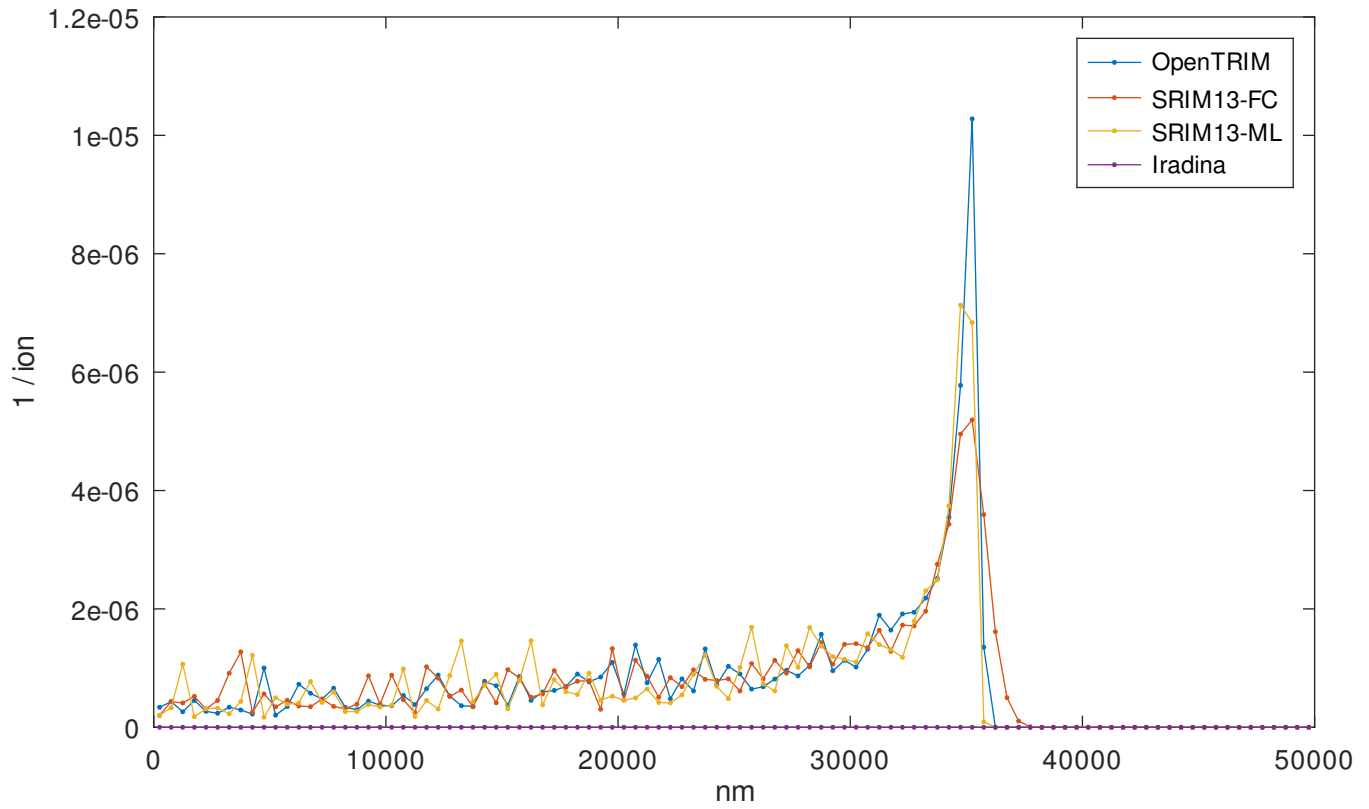
Implanted H ion



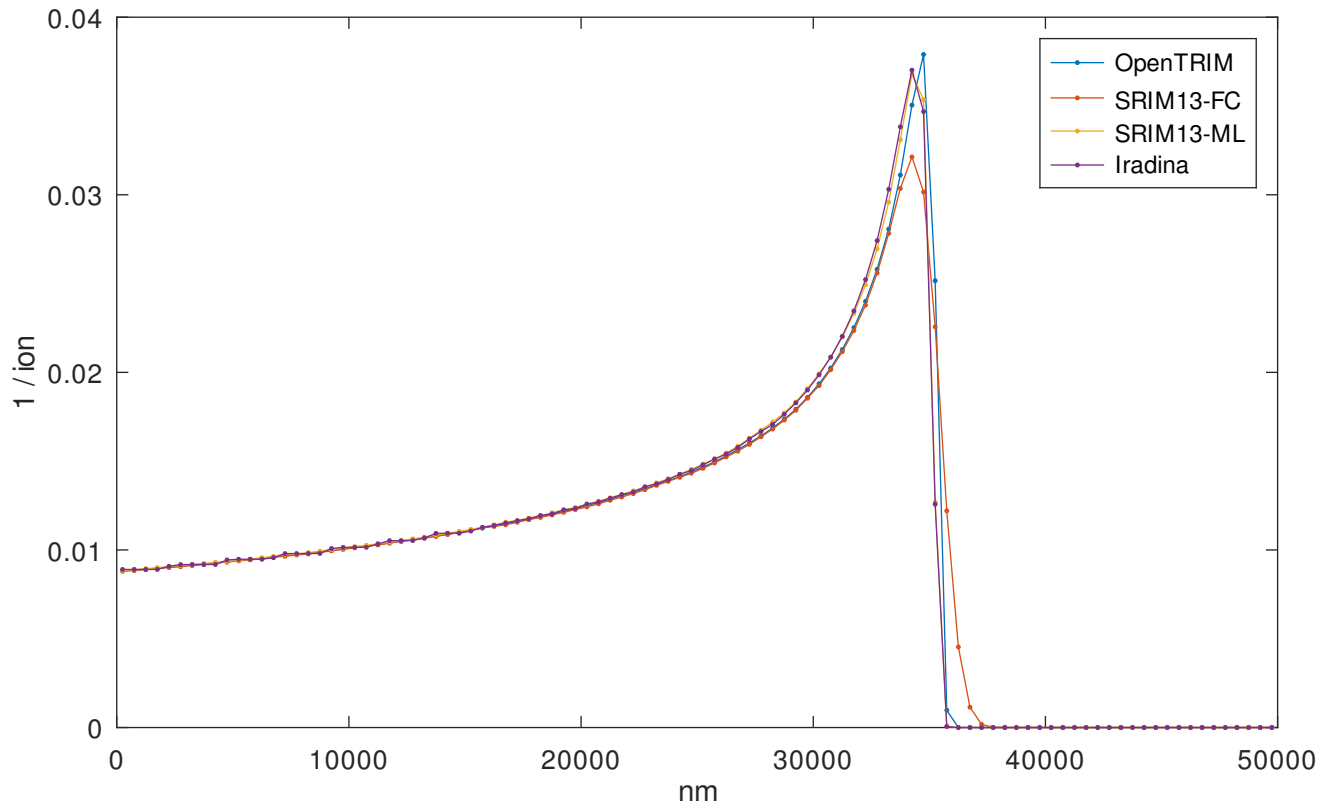
Electronic energy loss fraction $\Delta E_e/E_0$ by H ion



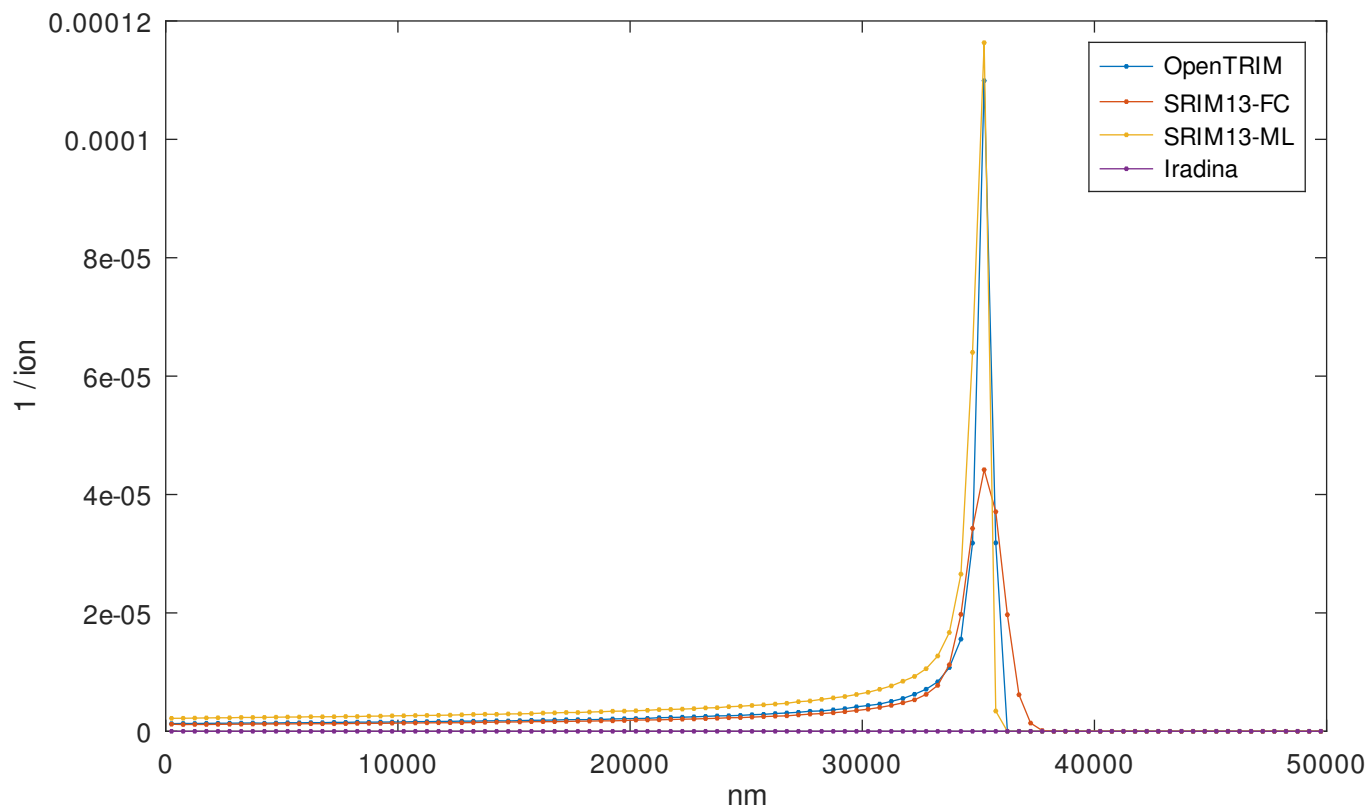
Electronic energy loss fraction $\Delta E_e/E_0$ by recoils



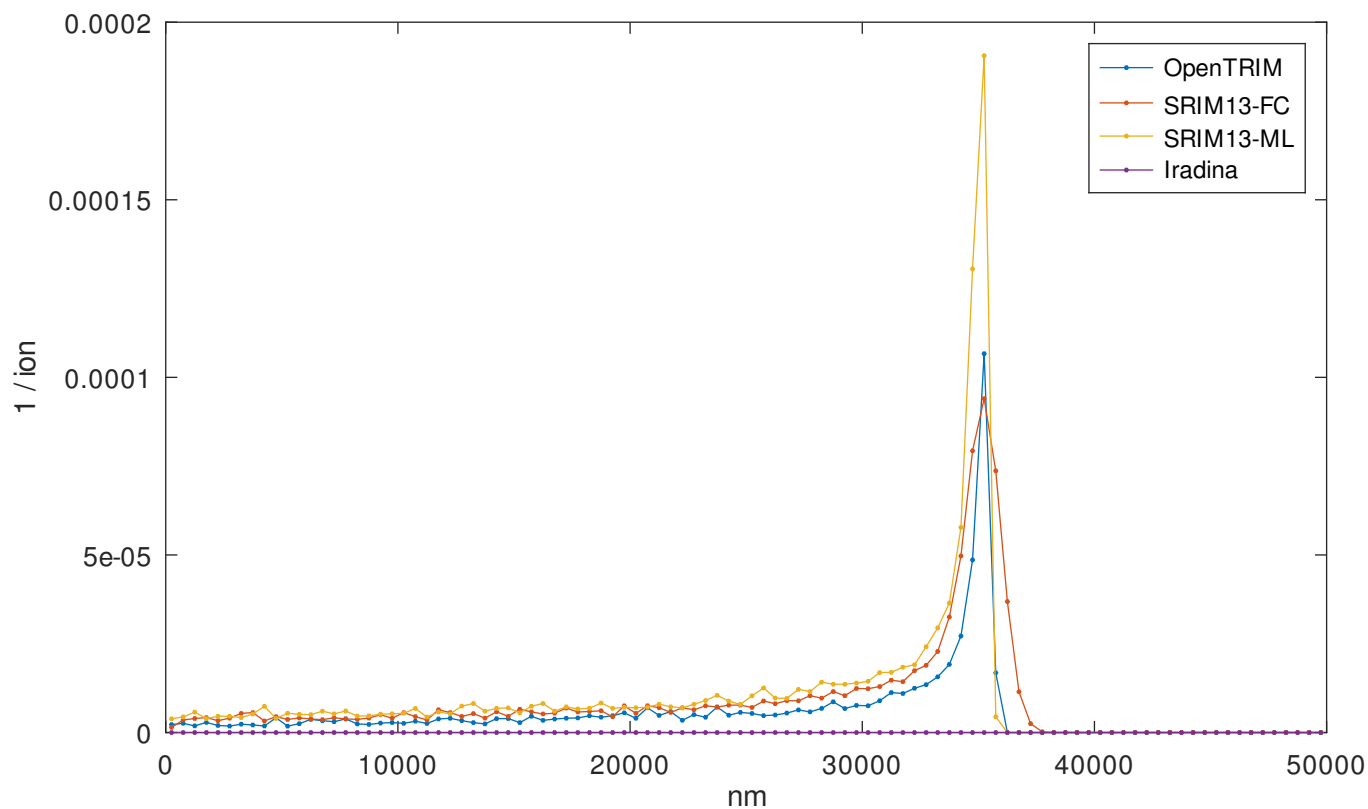
Total Electronic energy loss fraction $\Delta E_e/E_0$



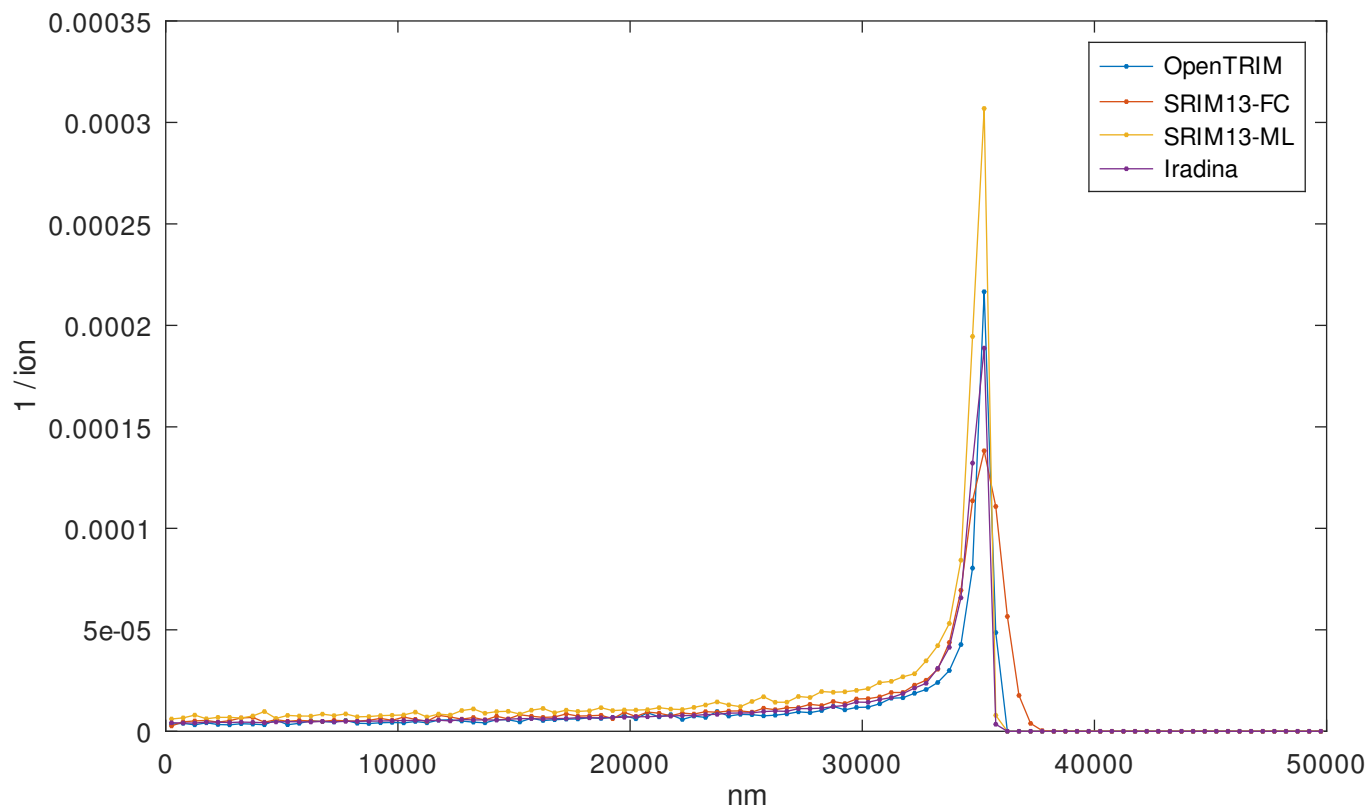
Phonon energy fraction $\Delta E_{Ph}/E_0$ by H ion



Phonon energy fraction $\Delta E_{Ph}/E_0$ by recoils



Total Phonon energy fraction $\Delta E_{Ph}/E_0$



Total fractional energy deposition $(\Delta E_e + \Delta E_{Ph})/E_0$

